

Final Project: Financial News Sentiment & FDIC Banking Stress

Project Information

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Overview

This project asks:

1. Can LLM-derived sentiment signals from financial news predict deteriorations in state-level banking fundamentals (FDIC)?
2. Do sentiment indicators provide earlier or stronger signals than traditional FDIC fundamentals (e.g., ROA/NIM)?
3. Are stress reactions geographically concentrated across US states?

Research Questions

This project studies whether regulatory-finance news sentiment contains useful early warning signals for banking stress at the US state-year level.

1. **Predictive signal:** Is a higher share of negative regulatory-news sentiment associated with worse banking outcomes (especially lower next-period ROA)?
2. **Incremental value:** Does sentiment improve stress identification when used together with FDIC fundamentals such as lagged ROA, NIM, and balance-sheet growth?
3. **Spatial concentration:** Are model-implied stress levels clustered in particular states, suggesting uneven geographic vulnerability?

Data

- News/regulatory text: `raw_partner_headlines.csv` (filtered to regulatory keywords; scored with a lightweight sentiment model)
- Banking fundamentals: FDIC state-year aggregates (e.g., `Summary_data_states.csv`)
- Geography: US state boundaries (Census shapefile; used for choropleth map)

Folder conventions

- Raw inputs go in `data/raw-data/`
- Derived/clean tables go in `data/derived-data/`

Data & Methods

All of the processing, merging, and reshaping for this project is handled directly in this `.qmd` file via Python code chunks. The helper module `preprocessing.py` exposes reusable functions (for presentation slides and the Streamlit app), but the core wrangling logic is executed here.

Method summary:

1. Load FDIC state-year data and construct derived features such as `ROA`, `DROA`, lagged controls, and stress labels (`bad_year`, `severity`).
2. Train a lightweight TF-IDF + logistic-regression sentiment model using the labeled financial text corpus.
3. Score filtered regulatory-news headlines and aggregate sentiment to the year level.
4. Merge yearly sentiment with FDIC panel data.
5. Construct a simple composite stress index `StressScore = bad_year × severity` for visualization.

Build state-year panel from raw data

Load FDIC state-year aggregates from `Summary_data_states.csv` and construct ROA, Δ ROA, and related fundamentals.

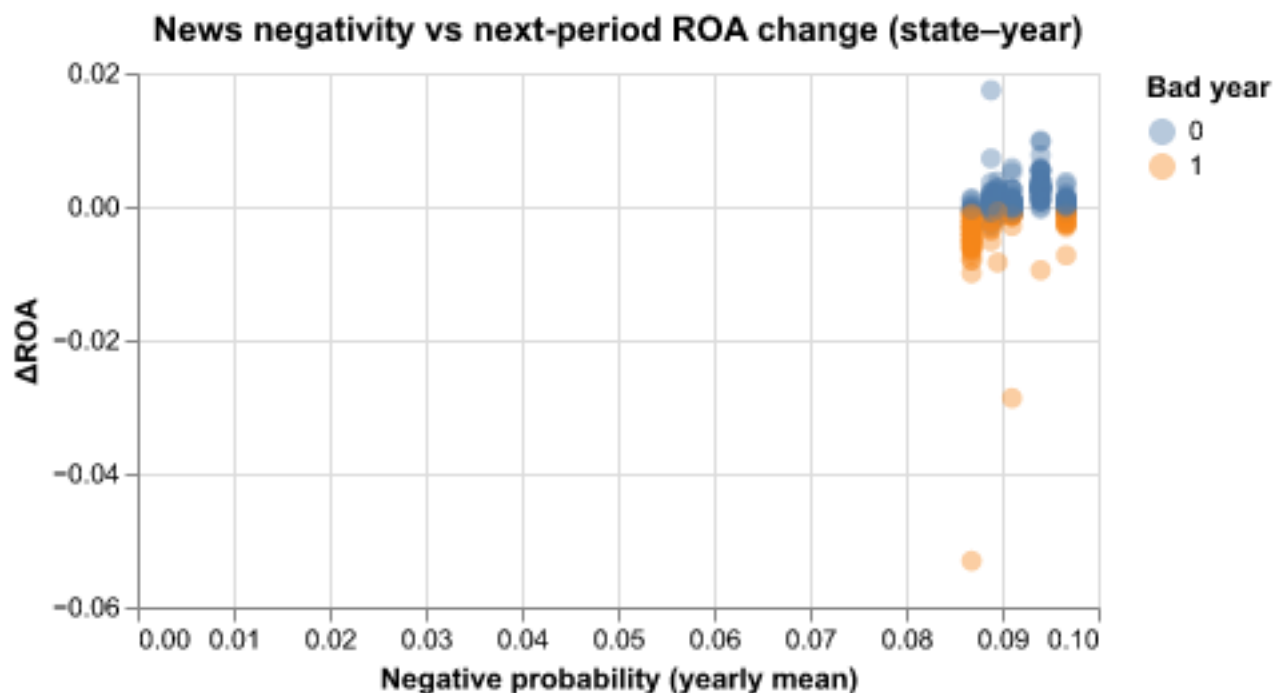
Train a lightweight sentiment model on the labeled news dataset and aggregate headline sentiment to the year level. To make this chunk self-contained (so it works even if run independently), we reload the FDIC panel to determine the relevant year range.

Merge the FDIC panel with the yearly sentiment index and construct a simple composite stress index `StressScore` that combines the binary bad-year label with the magnitude of ROA deterioration.

Results

Static Plot 1: Sentiment vs Δ ROA scatter

We first show how regulatory-news negativity relates to changes in ROA at the state-year level using an Altair scatter plot.



Interpretation: higher `sent_neg_share` generally aligns with weaker `ΔROA`, supporting Research Questions 1 and 2.

Static Plot 2 (Spatial): US map of stress indicators

Next we map the `StressScore` across US states using Altair. We focus on the most recent year available in the panel.



Interpretation: darker states indicate higher stress, showing geographic concentration relevant to Research Question 3.

Streamlit App

The Streamlit app lives in `streamlit-app/`. See `README.md` for how to run it locally. The deployed version is available on Streamlit Community Cloud at <https://final-project-ppha30538-r9vojv7vmmmqe82uikv.streamlit.app/>.

The app is designed to support the writeup in three ways:

- **Exploration:** users can inspect sentiment and banking-stress indicators interactively instead of relying only on static snapshots.
- **Cross-checking:** users can compare patterns seen in the static figures with additional years or states in an interactive view.
- **Communication:** the app provides an accessible interface for explaining how sentiment features and FDIC fundamentals jointly inform the stress signal.

Limitations

- The project guidelines discourage state-level analysis in favor of more disaggregated data. We use state-year aggregation here because the FDIC “Summary of Results” and publicly available regulatory-news datasets are most readily available at this level; finer-grained (e.g., bank-level) data would require different sources and access.
- The headline source used for yearly sentiment may be incomplete in this repository snapshot; when missing, sentiment aggregates are unavailable.
- The modeling strategy is intentionally lightweight and does not establish causal effects between sentiment and banking outcomes.
- State-year aggregation can hide within-state bank heterogeneity and timing effects at higher frequency (quarterly/monthly).

Policy implications

- **Early-warning context:** **StressScore** should be used as a monitoring indicator to prioritize supervisory attention, not as a stand-alone forecast.
- **Targeted response:** when both fundamentals and news sentiment deteriorate in the same states, policymakers can prioritize region-specific interventions.